Is LLM self-consistency a usable signal for routing annotation work? Project Proposal for NLP Course, Winter 2026/27

Tymoteusz Urban
320665
01171272@pw.edu.pl

Sebastian Trojan
320664
01171271@pw.edu.pl

Mateusz Nizwantowski
313839
01161932@pw.edu.pl

supervisor: Soveatin Kuntur
Warsaw University of Technology
soveatin.kuntur.dokt@pw.edu.pl

Abstract

Large language models (LLMs) are increasingly used to annotate text, but repeated agreement between model runs is useful for annotation routing only if it predicts correctness. This project tests whether self-consistency across five reruns can identify correct multilabel annotations of persuasion and manipulation techniques across English and Polish. We will evaluate Bielik, PLLuM and Gemma 4 on three data sources: Polish SemEval-2023 Task 3 paragraphs and Polish MIPD articles, with English SemEval-2023 Task 3 paragraphs as an additional source. Across language-appropriate prompt strategies, we will compare consistency with token log-probability, verbalised confidence and random routing. Unlike work that treats rerun variability as an outcome in itself, we assess whether agreement is diagnostic of error and whether it supports a practical cost–quality trade-off. Using held-out evaluation and calibration-only threshold selection, we will measure predictive performance, annotation coverage, quality, human review effort and inference cost, with uncertainty estimates. The project will deliver a reproducible evaluation pipeline, prompts and logging code, alongside consistency–correctness analyses and cost–quality routing curves. Repeated inference will use a performance-optimised vLLM setup, with runtime and GPU memory tracked. The study may establish that self-consistency is a useful routing signal or provide evidence that it offers no practical advantage over simpler alternatives.

1 Introduction

1. **Scientific goal of the project.** This study asks whether self-consistency across LLM reruns predicts correctness of multilabel annotations of persuasion and manipulation techniques in English and Polish texts and supports cost-effective human–model routing. H1: higher self-consistency distinguishes correct from incorrect predictions better than chance across models, data sources and prompting strategies. H2: at comparable annotation quality, consistency-based routing achieves greater automatic coverage than alternative confidence-based approaches, including cheaper single-run methods. We also evaluate risk–coverage trade-offs, human annotation effort saved and additional inference costs. Negative results are considered valid research outcomes.

2. **Significance of the project.** Annotation is costly, but LLMs can be consistently wrong. Repeated inference also adds compute cost, so improved routing must be compared with cheaper confidence signals. We evaluate self-consistency for multilabel text classification across English and Polish data, examining its practical cost–quality trade-off across languages and annotation tasks. Model comparisons are descriptive, not causal evidence about model size or language specialisation.

3. **Concept and work plan.** Figure 1 summarises the workflow.

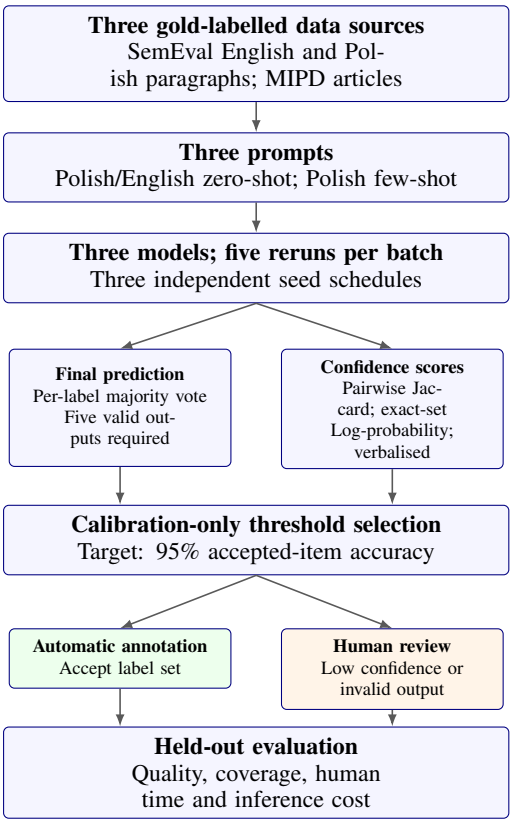


Figure 1: Planned annotation-routing pipeline. Thresholds are calibrated without test labels; human correction is simulated with gold labels.

Date	Planned output and owner
04 Nov	**Phase 1:** Methodology and pipeline (S); SOTA (T); datasets and data statement (M).
11 Nov	Docker/Conda/vLLM, GPU setup and data splits (M); inference pipeline and parsing (S); prompt preparation (T).
18 Nov	**Mid-term:** Three-model pilot with five reruns (S,M); SOTA presentation and video (T); start main runs (M).
25 Nov	Continue main inference (M); consistency metrics and confidence baselines (S); EDA and annotation-time sample (T).
02 Dec	**Phase 2:** Main inference and reproducible MR/README (M); preliminary evaluation (S); report and rebuttals (T).
09 Dec	Validate outputs and recover failures (M); AUC and confidence comparisons (S); preliminary error analysis (T).
16 Dec	Routing, calibration and bootstrap (S); GPU performance measurements (M); 20+ classified errors (T).
23 Dec	Risk–coverage and cost analysis (S); results discussion (T); reproducibility tests (M).
30 Dec	Complete report draft (T); final figures and results (S); code and README audit (M).
06 Jan	Final results verification (S); report revision and presentation video (T); final merge request (M).
13 Jan	**Phase 3:** Final report, presentation, code and MR (S,T,M).

Table 1: Weekly plan for 2026/27. S = Sebastian Trojan; T = Tymoteusz Urban; M = Mateusz Nizwantowski. All dates are Wednesdays; January dates refer to 2027.

Three deliverables. (i) Consistency–correctness analyses per model, dataset and supported label, with uncertainty estimates and confidence baselines. (ii) Risk–coverage and cost–quality curves, including comparison with single-run routing and a selected operating point (if feasible). (iii) Reproducible prompts, code, settings and outputs; sensitive outputs remain private.

Risks and responses. Pilot runtime will determine a feasible shared sample and GPU settings. Audit data/model licences, missing labels and document leakage. Report rare-label limitations; route invalid outputs to humans. Record uncertainty and examine at least 20 misclassified examples grouped by error type.

Timeline and ownership. Table 1 presents weekly tasks, milestones and individual responsibilities.

4. **Approach and research methodology.** *Data and models.* Use the three sources described under Datasets, each with its native label taxonomy. Because SemEval test gold labels are unpublished, pool SemEval train and dev and re-split them by article. Models: Bielik 11B v3 Instruct (SpeakLeash), PLLuM 12B Chat 2412 (CYFRAGOVPL) and Gemma 4 E2B IT (Google).

Prompt strategies. Use Polish zero-shot, English zero-shot and Polish few-shot prompts with equivalent instructions, label IDs and output schemas. Few-shot examples come only from training data; comparisons isolate instruction language or demonstrations, not a full factorial design.

Repeated inference and failures. Generate five outputs per model, item and prompt using fixed stochastic settings and logged seeds. Replicate the main comparison using three

independent five-output batches on a common pilot-sized sample, reporting the spread. Log raw and parsed outputs, versions and settings. Refer items with any invalid output to a human; report failures separately from classification accuracy and retain all items in coverage denominators.

Prediction and self-consistency. For five valid outputs, predict each label by majority vote. Measure exact-set agreement (modal-set frequency), mean pairwise Jaccard similarity over ten pairs (primary score), and per-label agreement. Confidence scores never use gold labels.

Confidence baselines. On the same voted predictions, compare self-consistency with length-normalised token log-probability of the canonical voted label set, scored by teacher forcing under the original prompt; verbalised confidence elicited on a fixed $[0, 1]$ scale; and random routing. Log additional scoring/query costs. Report unavailable log-probability baselines explicitly.

Metrics and uncertainty. Report micro/macro-F1 and exact-set accuracy on valid predictions, plus failure rates. Evaluate ROC AUC for predicting exact-set correctness, supplemented by error-detection PR AUC. Analyse per-label correctness only with adequate support. Use paired document-level bootstrap 95% confidence intervals and report spread across independent seed schedules.

Routing evaluation. Calibrate thresholds without test data, maximising coverage subject to 95% accepted-item exact-set accuracy and a predeclared minimum accepted count; also examine 80% and 90% targets. Evaluate risk–coverage curves, accepted-only quality and final quality under ideal gold-based human correction on held-out data. Report infeasible targets instead of assuming coverage.

Cost–benefit analysis. Compare confidence signals on matched five-run predictions and complete policies against cheaper single-run baselines. Report coverage, quality, inference time and estimated human hours based on a timed sample from each dataset. Estimate net monetary cost using explicit labour and compute rates, including reruns and confidence queries; do not assume savings.

Tools. Use Python and a performance-optimised vLLM configuration on local NVIDIA GPUs, with a Dockerfile and Conda environment specification. Optimise inference throughput and memory use while preserving the fixed decoding settings required for reproducibility. Log versions, model settings, seeds, GPU memory, runtime and execution commands. No model fine-tuning is planned.

5. **State of the art.**

   Our work is related to three research areas: detection of persuasion and manipulation techniques, confidence estimation for LLM outputs, and LLM-assisted annotation. Table 2 lists published numbers of 15 selected works with the dataset and split for each of them.

   Persuasion and manipulation detection. On SemEval-2023 Task 3 (Piskorski et al., 2023), which is our English and Polish source, the best fine-tuned models reach 0.376 test micro-F1 for English and 0.430 for Polish (Purificato and Navigli, 2023; Hromadka et al., 2023), while macro-F1 stays below 0.18, so rare techniques are rarely detected. On MIPD (Modzelewski et al., 2024), fine-tuned Polish encoders reach a weighted F1 of 0.47. Prompted LLMs mostly have worse F1 scores than fine-tuned encoders, but detect rare persuasion and manipulation techniques better (Hasanain et al., 2024) - a zero-shot chain-of-thought GPT-4o-mini prompt won a recent Polish persuasion shared task (Jose and Greenstadt, 2025). We found no evaluation of open Polish LLMs on these tasks.

   Confidence of LLM outputs. Majority voting on outputs improves accuracy (Wang et al., 2023), and consistency among samples detects hallucinations better than token log-probabilities (Manakul et al., 2023). Verbalised confidence (e.g. "I am sure for 80%") is also often better calibrated than token probabilities (Tian et al., 2023). However for text classification consistency over four samples was the weakest of these three methods (consistency, verbalised, token log-prob), partly because its scores take few distinct values (Merdjanovska et al., 2026)

Work	Task / method	Data (language), split	Model	Result
Persuasion and manipulation detection				
Piskorski et al. (2023)	Shared task: multilabel detection of 23 techniques in paragraphs	SemEval-2023 T3 ST3 (EN, PL), official test	SVM baseline	micro-F1: EN 0.195, PL 0.179
Purificato and Navigli (2023)	Ensemble of five fine-tuned encoders	SemEval-2023 T3 ST3 (EN, PL), official test	APatt	micro-F1: EN 0.376 (1st place), PL 0.366
Hromadka et al. (2023)	One model fine-tuned on all languages	SemEval-2023 T3 ST3 (PL, EN), official test	XLM-R-large	micro-F1: PL 0.430 (1st place), EN 0.362
Modzelewski et al. (2024)	Multilabel detection of 11 manipulation techniques in articles	MIPD (PL), test, 1521 articles	HerBERT-large	weighted F1 0.47 (mean of 5 seeds)
Hasanain et al. (2024)	Prompted GPT-4 compared with fine-tuned encoders	ArPro (AR), test	XLM-R, GPT-4	micro-F1: XLM-R 0.608, GPT-4 0.467; macro-F1: XLM-R 0.128, GPT-4 0.212
Jose and Greenstadt (2025)	Zero-shot prompting, 25 techniques	SlavicNLP 2025 subtask 2 (PL), official test	GPT-4o-mini	micro-F1 0.419 (1st place), macro-F1 0.296
Confidence of LLM outputs				
Wang et al. (2023)	Majority vote over 40 sampled answers	GSM8K (EN), test	PaLM-540B	accuracy 0.744 (single answer: 0.565)
Manakul et al. (2023)	Agreement between samples to detect hallucinations	WikiBio GPT-3 (EN), all 1,908 sentences	GPT-3	AUC-PR 0.934 (token log-prob: 0.832)
Tian et al. (2023)	Model states its own confidence	TriviaQA (EN), validation, 1000 questions	GPT-3.5-turbo	ECE reduced by a relative 50% in many settings
Merdjanovska et al. (2026)	Compares consistency, stated confidence and log-prob for classification	SST-2 and SST-5 test, Yahoo and ESCI samples (EN), average	Qwen3-32B	AUROC: consistency 0.573, stated 0.661, log-prob 0.623
Geifman and El-Yaniv (2017)	Model may abstain below a confidence threshold	ImageNet, half of validation set held out	ResNet-50	top-5 error 0.019 at coverage 0.594
LLMs as annotators				
Gilardi et al. (2023)	Zero-shot annotation compared with crowd workers	6183 tweets and news articles (EN)	ChatGPT	accuracy about 0.25 higher than MTurk
Atıl et al. (2025)	Ten runs with temperature 0 and fixed seed	MMLU college math (EN), 100 questions	GPT-4o	median accuracy 0.69
Li et al. (2023)	Splits items between LLM and humans by answer entropy	TempoWiC (EN), test, 40% of items labelled by LLM	ChatGPT (annotation), RoBERTa (classification)	macro-F1 0.569 (random split: 0.532)
Pangakis and Wolken (2025)	Consistency over at least 5 reruns	27 tasks (EN), 1000 human-labelled items each	GPT-4	accuracy with consistent labels 0.194 higher than the rest

Table 2: Published results of the 15 reviewed works. ST3 = subtask 3, ECE = expected calibration error.

and verbalised and token log-probabilities give continuous-valued scores. We include work Geifman and El-Yaniv (2017), which introduces selective classification and risk-coverage terms for deep networks.

LLMs as annotators. LLMs can outperform crowd workers (Gilardi et al., 2023), but still their accuracy varies across reruns even with set seeds and settings (Atıl et al., 2025). Most similar to our work, Pangakis and Wolken (2025) rerun GPT-4 at least 5 times on 27 binary tasks and found that fully consistent labels are almost 20 percentage points more accurate than the rest.

Space for research. Evidence on rerun agreement comes from English with binary or single-label tasks, but no study checks how many items can be labelled automatically for a fixed accuracy (e.g. 95%) across different routing methods (verbalised confidence, token log-probabilities, consistency). We test this for multilabel persuasion and manipulation annotation in Polish and English with open models.

6. **Datasets.** All three sources are READY: publicly available and downloaded; dataset sizes are summarised in Table 3. Label definitions and label frequencies are provided in Appendix A (Table 5; Figure 2). Polish data are primary; English SemEval is an additional source for cross-lingual comparison.

SemEval-2023 Task 3 (Piskorski et al., 2023) was created for a multilingual shared task on news genre, framing and persuasion techniques. It contains news and web articles (2020–mid 2022) from mainstream and alternative outlets on topics such as COVID-19, climate change, migration and the war in Ukraine. We use subtask 3: paragraph-level multilabel detection of 23 persuasion techniques grouped into six categories; only 19 occur in the English data. We downloaded the English and Polish data from the UIUC team's GitHub mirror[1]; it is used for research purposes only. Test gold labels are not public, so only train and dev are used; 16 Polish label rows without matching text are dropped.

MIPD (Modzelewski et al., 2024) was created to study manipulation and intention in Polish online disinformation. It contains

[1] https://github.com/yrf1/SemEval23-Task-3-UIUC-Team

15,356 web articles (mean 767 words) from more than 400 public sources, including government sites, alternative media and conspiracy or propaganda websites. Five professional fact-checkers with at least three years of experience annotated the articles; two experts assessed each article, disagreements were resolved by discussion, and unresolved "hard to say" cases were excluded. Besides credibility, intention and thematic categories, articles carry 11 manipulation-technique labels, the only labels we use. We keep all articles; credible articles have empty label sets (76%). The dataset is distributed on GitHub[2] under CC BY-NC-ND 4.0.

Source (unit)	Docs	Items	Labels	Med. words	Empty
SemEval EN (paragraph)	536	12,625	19/23	28	61%
SemEval PL (paragraph)	194	3,094	23/23	39	47%
MIPD PL (article)	15,356	15,356	11/11	544	76%

Table 3: Dataset sizes. SemEval: train+dev (EN 446+90, PL 145+49 articles); MIPD: official train/validation/test (10,749/3,086/1,521). Labels = techniques present / taxonomy size; Empty = items with no technique.

7. **Data statement.** SemEval-2023 Task 3 EN/PL (research use) and MIPD (CC BY-NC-ND 4.0) are used non-commercially with attribution. All texts are publicly available news and web articles. They contain no personal data beyond public figures named in news reporting, and we collect none. We do not attempt to identify authors, modify the datasets or fine-tune models on them.

8. **Contribution table.** Table 4 records Phase 1 work.

Member	Phase 1 contribution	h
Tymoteusz Urban	State-of-the-art literature review.	8
Sebastian Trojan	Scientific goal, concept, methodology, plan, contribution table and LLM-use declaration.	5
Mateusz Nizwantowski	Repository and Overleaf setup; Datasets and status; data statement.	8
Total		21

Table 4: Phase 1 division of work.

[2] https://github.com/ArkadiusDS/MIPD

9. **Declaration of LLM use.** OpenAI Codex supported drafting, language editing and LaTeX formatting, including generating the abstract from the proposal body. The authors are responsible for the final content.

   Claude Code was used to help with finding adequate literature and SOTA solutions as well as to generate a table with summary (which was proof checked and edited).

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A Dataset details

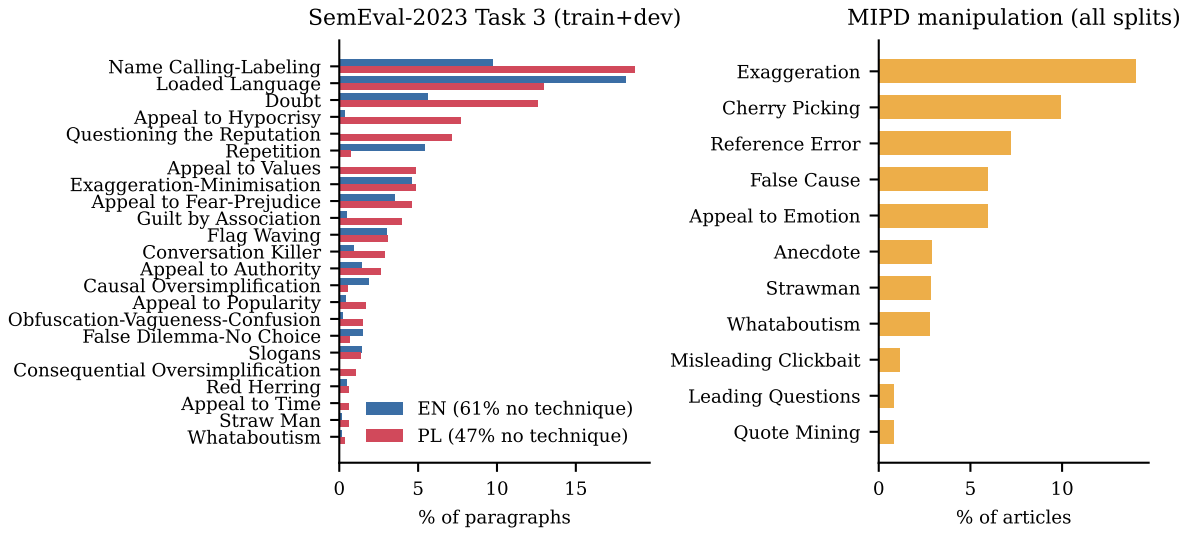


Figure 2: Label frequencies. Left: SemEval-2023 Task 3 train+dev (% of paragraphs; EN in blue, PL in red). Right: MIPD manipulation labels across all splits (% of all articles; 76% have no technique label).

SemEval-2023 Task 3 (paragraph level)				**MIPD** (article level)	
Justification		*Call*		Cherry Picking	Selective data supporting a claim
Appeal to Authority	Cites an authority as support	Slogans	Brief, striking phrase	Quote Mining	Distorts statements via selective excerpts
Appeal to Popularity	"Everybody agrees"	Appeal to Time†	The time to act is now	Anecdote	Personal stories against statistical evidence
Appeal to Values†	Links idea to positive values	Conversation Killer	Discourages further discussion	Whataboutism	Deflects via an unrelated argument
Appeal to Fear-Prejudice	Exploits fear or prejudice	*Manipulative wording*		Strawman	Attacks a misrepresented argument
Flag Waving	Appeals to group pride or benefit	Loaded Language	Emotionally charged words	Leading Questions	Questions implying a set conclusion
Simplification		Repetition	Repeats words or phrases to persuade	Appeal to Emotion	Manipulates readers' emotions
Causal Oversimplification	Assumes a single cause	Exaggeration-Minimisation	Over- or understates	False Cause	Causation from mere correlation
False Dilemma-No Choice	Only two options presented	Obfuscation-Vagueness-Confusion	Deliberately unclear wording	Exaggeration	Over- or understates facts
Consequential Oversimplification†	Unproven chain of consequences	*Attack on reputation*		Reference Error	Cites unreliable or false sources
Distraction		Name Calling-Labeling	Loaded labels for a target	Misleading Clickbait	Headline contradicting the article
Straw Man	Refutes a distorted argument	Doubt	Questions target's credibility		
Red Herring	Diverts to another topic	Guilt by Association	Links target to a disliked group		
Whataboutism	Counter-accusation instead of answer	Appeal to Hypocrisy	Charges target with hypocrisy		
		Questioning the Reputation†	Attacks character or morals		

Table 5: Label taxonomies with abridged definitions (SemEval: official guidelines, grouped by the six official categories; MIPD: dataset documentation). †Absent from English SemEval data.